Study Guide — Upayan Mazumder

Presenter focus: Central architecture
Registration: 24BDS0367
Use this guide to prepare your presentation section.

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1. Overview for the Presentation

You own the central architecture. Explain how the whole system works as one closed loop. Cover:

- System model and data flow
- CTC link (WBeacon / ZBeacon)
- SimPy simulation engine
- Shared 2.4 GHz channel medium
- Metrics that close the evaluation loop

Nitesh covers WiFi internals. Divyansh covers ZigBee internals. Your job is to join both sides and show the shared machinery.

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2. Quick Brief: Nitesh's WiFi Work

A Frame Group (FG) is a WiFi batch with an arrival time, duration, and deadline. The AP cycles through `MAINTAINING → CTC_GATHERING → RESCHEDULING → CTC_NOTIFYING → TRANSMITTING`.

In `RESCHEDULING`, the MILP tries to place a free window and a protected sensing window. It may leave WiFi fragments in the free window. The solver has a 10-second cycle limit. The AP uses whitespace expansion when the solver has no usable result.

Baselines:

- Exp 1: no coordination
- Exp 2: opportunistic idle-gap sensing
- Exp 3: ECC / NAV reservation with a fixed ZigBee superframe

If asked for MILP details, ask Nitesh to explain overlap, absolute movement, postponement, and deadline costs.

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3. Quick Brief: Divyansh's ZigBee Work

The ZigBee Coordinator (ZC) reports demand with a ZBeacon. After it receives a WBeacon, it runs L-WSA.

L-WSA searches over slot duration t_{SfSlot}, CAP slots i, CFP slots j, and beacon offset t_{offset}. Hard limits: $j \leq 7$ and $i + j \leq 15$. The search maximises usable CAP and CFP efficiency while avoiding residual WiFi fragments.

ZC states: `IDLE`, `CTC_REPORTING`, `COMMAND_ACQUISITION`, `TRAFFIC_ADJUSTMENT`, `SUPERFRAME_ACTIVE`.

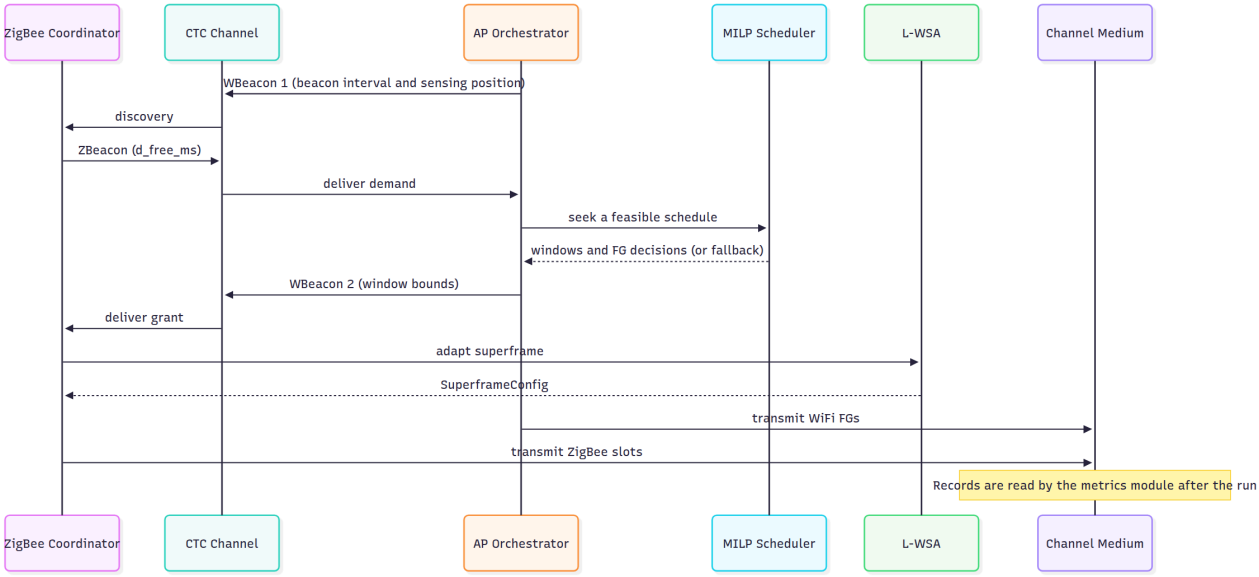
If asked for L-WSA search details, hand off to Divyansh.

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4. Deep Dive: Central Architecture

4.1 Paper handshake and simulated cycle

The paper starts with WBeacon 1. Our simulation models ZBeacon demand and a WBeacon grant. It does not transmit a separate WBeacon 1.



Paper steps in words (the first broadcast is not an event in our model):

1. The AP broadcasts WBeacon 1. The ZC learns when the AP senses for demand.
2. The ZC sends a ZBeacon with its demand in the sensing window.
3. The AP solves the WiFi schedule or uses the fallback.
4. The AP sends WBeacon 2 with the free-window and sensing-window bounds.
5. The ZC runs L-WSA and avoids WiFi fragments in the free window.
6. Both sides transmit into `ChannelMedium`. Overlapping bursts fail.
7. The metrics module reads the channel records after the run.

4.2 CTC channel model

Module: `src/cowic/ctc/channel.py`

Message	Direction	Payload
ZBeacon	ZC → AP	`d_free_ms`
WBeacon	AP → ZC	`t_free_start_ms`, `t_free_end_ms`, `t_sense_start_ms`, `t_sense_end_ms`

Notes:

- Delivery is discrete-event. There is no waveform emulation.
- Packet error rate (PER) is a configurable parameter.
- Base-paper PHY names: WiCast (downlink), SymBee (uplink). The software model abstracts them as typed control messages.

4.3 SimPy simulation engine

Module: `src/cowic/sim/simulation_engine.py`

Role:

- Drive AP and ZC processes on a shared SimPy environment
- Replay WiFi traces cycle by cycle
- Call orchestrator / coordinator methods at the right simulated times
- Collect channel records and pass them to metrics

The engine also synthesizes ZigBee attempts and WiFi airtime from the schedule. It handles carryover across cycles and applies the CTC result.

4.4 Shared channel medium

Module: `src/cowic/sim/channel_medium.py`

Role:

- Accept transmission intervals from WiFi and ZigBee
- Detect temporal overlap beyond `OVERLAP_TOLERANCE_MS`
- Mark collisions and successes
- Expose records for metrics

Collision rule in this project: temporal overlap on the shared channel causes mutual failure of the overlapping transmissions.

4.5 Models layer

`FrameGroup` (WiFi): identity, arrival, duration, deadline, original start, QoS flag.

`SuperframeConfig` (ZigBee): `slot_duration_ms`, `cap_slots`, `cfp_slots`, `beacon_offset_ms`, `beacon_interval_ms`.

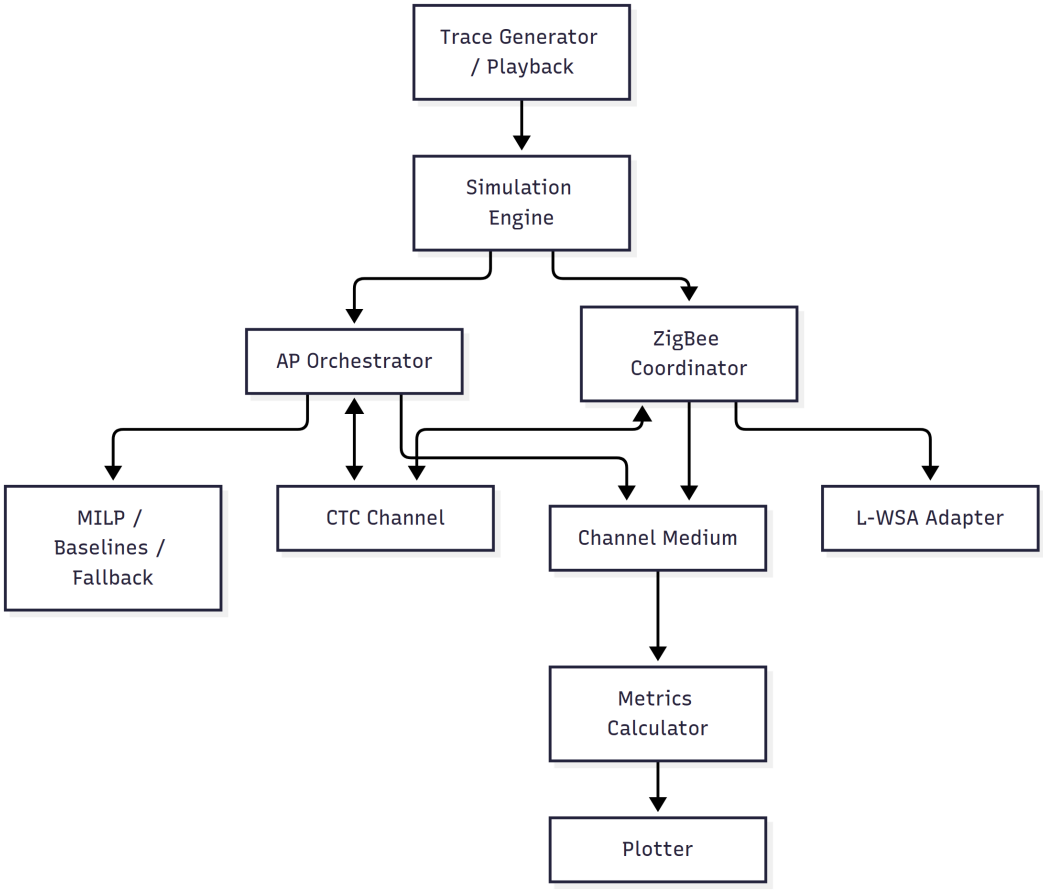
4.6 Metrics

Module: `src/cowic/analysis/metrics.py`

Metric	Meaning
G_z	Delivered ZigBee payload / ideal clean-spectrum payload over the same demand.
PDR_w	Lost WiFi FG airtime / requested FG airtime. Loss includes deadline drops and collisions.
U_{eff}	Successful useful airtime / simulated duration.
Jitter	Variation in WiFi FG delay.
Max delay	Longest WiFi FG delay.

The 60-second Traffic I run gives Exp 4 G_z of 0.94, 0.87, and 0.63 at D_{free} of 31, 62, and 123 ms. Its PDR_w is 1.3 %, 2.7 %, and 4.0 %. Do not claim zero WiFi loss on this trace.

4.7 Full architecture diagram



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5. Likely Questions and Answers

Q1. Why software-only?
A. The paper evaluates a MAC protocol in simulation. Our SimPy model studies timing and collisions, not waveforms or radio hardware.

Q2. What does CTC carry in this project?
A. Control scalars only. ZBeacon carries demand. WBeacon carries free and sensing window bounds.

Q3. How do you detect a collision?
A. `ChannelMedium` compares time intervals. Overlap beyond tolerance marks both sides as collided.

Q4. Where does the sensing window sit?
A. The AP places it. The MILP hard constraint forces zero WiFi overlap with the sensing window.

Q5. What happens if MILP fails?
A. The AP uses whitespace expansion. The fallback can still lose FGs or grant less ZigBee airtime under heavy load.

Q6. How does the engine keep WiFi and ZigBee aligned?
A. Both processes run in one SimPy environment. CTC messages and cycle boundaries synchronise them.

Q7. What is residual WiFi fragmentation?
A. Some WiFi FGs cannot move. They leave fragments inside the free window. L-WSA must place ZigBee slots around them.

Q8. How is G_z computed?
A. Divide delivered ZigBee payload by the ideal payload for the same demand under clean-spectrum conditions.

Q9. How do Exp 5 and Exp 6 help?
A. They are ablations. Exp 5 clears the window rigidly and uses fixed ZigBee slots. Exp 6 omits sensing overhead. Exp 5 reaches full ZigBee gain but loses up to 8.1 % of WiFi airtime on Traffic I.

Q10. What should the audience remember from your section?
A. The AP and ZC coordinate in one cycle. The model still loses WiFi airtime on high-load traces. The paper's absolute results do not all reproduce.